**R13**

**Code No: 118ET**

**JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD**
**B. Tech IV Year II Semester Examinations, July - 2023**
**STORAGE AREA NETWORKS**
**(Computer Science and Engineering)**

**Time: 3 Hours** **Max. Marks: 75**

**Note:** i) Question paper consists of Part A, Part B.
ii) Part A is compulsory, which carries 25 marks. In Part A, Answer all questions.
iii) In Part B, Answer any one question from each unit. Each question carries 10 marks and may have a, b as sub questions.

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## PART – A

**(25 Marks)**

| | | |
|---|---|---|
| 1.a) | What are the benefits of storage networks on business applications? | [2] |
| b) | What are the challenges in data storage and data management? | [3] |
| c) | Write the benefits of the different networked storage options. | [2] |
| d) | Discuss about RAID 0+1. | [3] |
| e) | Explain about business continuity. | [2] |
| f) | What are the reasons for planned outages? | [3] |
| g) | Explain about SAN based backup topology. | [2] |
| h) | Discuss about Network-based Remote Replication. | [3] |
| i) | What is a storage virtualization? | [2] |
| j) | What are the key characteristics of a data center? | [3] |

## PART – B

**(50 Marks)**

2. Explain the major physical components of a disk drive and their functions. [10]

**OR**

3. What are the limitations of server-centric IT architecture? How to overcome? Explain.[10]

4.a) Explain about SAN Connectivity.
b) Explain the topologies of FC-SAN. [5+5]

**OR**

5. Explain the following RAID levels with necessary block diagrams
a) RAID 3 b) RAID 5. [5+5]

6.a) Explain the reasons for unplanned outages.
b) Write a short note on RTO. [5+5]

**OR**

7. What are the single points of failure in a storage infrastructure and also list solutions to mitigate these failures? [10]

8.a) Discuss about LAN based backup topology.
b) Write a short note on Storage Array-based Remote Replication. [5+5]

**OR**

9.a) Explain about the Direct- attached Backup topology.
b) Write a short note on three-Site -based Remote Replication. [5+5]

10.a) List and Analyze the common threats in storage security domain.
b) Briefly explain about the critical security attributes for information systems. [5+5]

**OR**

11. Explain about block-level virtualization technologies and processes. [10]

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